

Step 1 — Install and Load spaCy Model

```
import sys

!{sys.executable} -m pip install spacy
!{sys.executable} -m spacy download en_core_web_sm

import spacy

nlp = spacy.load("en_core_web_sm")

print("spaCy model loaded successfully")
```

```
Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages (3.8.11)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.5)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.15)
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.13)
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (8.3.10)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.1.3)
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.5.2)
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.10)
Requirement already satisfied: weasel<0.5.0,>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.4.3)
Requirement already satisfied: typer-slim<1.0.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.24.0)
Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (4.67.3)
Requirement already satisfied: numpy>=1.19.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.2)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.32.4)
Requirement already satisfied: pydantic!=1.8,!1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.11.10)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.1.6)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from spacy) (75.2.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (26.0)
Requirement already satisfied: annotated-types==0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!1.8.1,<3.0.0,>=1.7.4) (0.6.0)
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!1.8.1,<3.0.0,>=1.7.4) (2.41.4)
Requirement already satisfied: typing-extensions==4.14.1 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!1.8.1,<3.0.0,>=1.7.4) (4.14.1)
Requirement already satisfied: typing-inspection==0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!1.8.1,<3.0.0,>=1.7.4) (0.4.2)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0) (3.4.1)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0) (3.10.1)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0) (2.3.1)
Requirement already satisfied: certifi<2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0) (2025.11.12)
Requirement already satisfied: blis<1.4.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4) (1.4.0)
Requirement already satisfied: confection<1.0.0,>=0.0.1 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4) (0.0.4)
Requirement already satisfied: typer==0.24.0 in /usr/local/lib/python3.12/dist-packages (from typer-slim<1.0.0,>=0.3.0) (0.24.0)
Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.4.2) (0.4.2)
Requirement already satisfied: smart-open<8.0.0,>=5.2.1 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.4.2) (7.0.5)
Requirement already satisfied: MarkupSafe==2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2) (2.0.1)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart-open<8.0.0,>=5.2.1) (1.17.0)
Requirement already satisfied: click==8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer==0.24.0) (8.2.1)
Requirement already satisfied: shellingham==1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer==0.24.0) (1.3.0)
Requirement already satisfied: rich==12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer==0.24.0) (12.3.0)
Requirement already satisfied: annotated-doc==0.0.2 in /usr/local/lib/python3.12/dist-packages (from rich==12.3.0) (0.0.2)
Requirement already satisfied: markdown-it-py==2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich==12.3.0) (2.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich==12.3.0) (2.19.0)
Requirement already satisfied: mdurl==0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py==2.2.0) (0.1.2)
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-any.whl: 12.8/12.8 MB 55.4 MB/s eta 0:00:00
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in order to load all the package's dependencies. You can do this by selecting the 'Restart kernel' or 'Restart runtime' option.
spaCy model loaded successfully
```

Step 2 — Sentence Processing

```
sentences = [
    "The cat sits on the mat.",
    "John bought a new car.",
    "The teacher explained the lesson clearly.",
    "Students are studying for the exam.",
    "Artificial Intelligence is transforming industries.",
    "Did you finish the assignment?",
    "The book was written by the author.",
```

```

"The quick brown fox jumps over the lazy dog.",
"She enjoys reading novels.",
"The project was completed successfully."
]
for sentence in sentences:
    doc = nlp(sentence)
    print("Sentence:", sentence)
    for token in doc:
        print(token.text, token.pos_, token.dep_)
    print("\n")

```

```

Sentence: The cat sits on the mat.
The DET det
cat NOUN nsubj
sits VERB ROOT
on ADP prep
the DET det
mat NOUN pobj
. PUNCT punct

```

```

Sentence: John bought a new car.
John PROPN nsubj
bought VERB ROOT
a DET det
new ADJ amod
car NOUN dobj
. PUNCT punct

```

```

Sentence: The teacher explained the lesson clearly.
The DET det
teacher NOUN nsubj
explained VERB ROOT
the DET det
lesson NOUN dobj
clearly ADV advmod
. PUNCT punct

```

```

Sentence: Students are studying for the exam.
Students NOUN nsubj
are AUX aux
studying VERB ROOT
for ADP prep
the DET det
exam NOUN pobj
. PUNCT punct

```

```

Sentence: Artificial Intelligence is transforming industries.
Artificial PROPN compound
Intelligence PROPN nsubj
is AUX aux
transforming VERB ROOT
industries NOUN dobj
. PUNCT punct

```

```

Sentence: Did you finish the assignment?
Did AUX aux
you PRON nsubj
finish VERB ROOT
the DET det
assignment NOUN dobj
? PUNCT punct

```

```

Sentence: The book was written by the author.

```

▼ Step 3 — Extract Dependency Relations

```

import pandas as pd

sentence = "The quick brown fox jumps over the lazy dog."
doc = nlp(sentence)

data = []

for token in doc:
    data.append([

```

```
data.append([
    token.text,
    token.lemma_,
    token.pos_,
    token.dep_,
    token.head.text
])

df = pd.DataFrame(data, columns=["Token", "Lemma", "POS", "Dependency", "Head Word"])
print(df)
```

	Token	Lemma	POS	Dependency	Head Word
0	The	the	DET	det	fox
1	quick	quick	ADJ	amod	fox
2	brown	brown	ADJ	amod	fox
3	fox	fox	NOUN	nsubj	jumps
4	jumps	jump	VERB	ROOT	jumps
5	over	over	ADP	prep	jumps
6	the	the	DET	det	dog
7	lazy	lazy	ADJ	amod	dog
8	dog	dog	NOUN	pobj	over
9	.	.	PUNCT	punct	jumps

Step 4 — Identify Sentence Structure

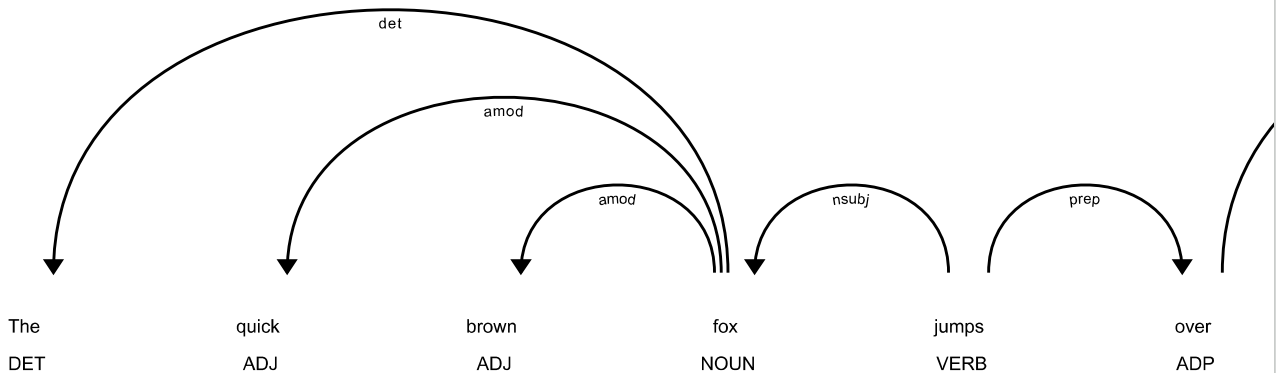
```
A=["The quick brown fox jumps over the lazy dog."]
```

Step 5 — Dependency Tree Visualization

```
from spacy import displacy

sentence = "The quick brown fox jumps over the lazy dog."
doc = nlp(sentence)

displacy.render(doc, style="dep", jupyter=True)
```



Step 6 — Comparative Analysis

Simple sentences produced short dependency trees with fewer branches.

Complex sentences contained multiple clauses, resulting in deeper trees.

Adjective modifiers like quick and lazy added extra nodes to the tree.

Prepositional phrases such as over the dog expanded the structure.

Passive sentences changed dependencies where the object becomes the subject.

For example, in passive voice nsubjpass appears instead of nsubj.

Sentences with conjunctions created additional dependency relations.

Dependency trees help clearly identify grammatical structure.

▼ Step 7 —Lab Report

1. Objective

To perform syntactic analysis using spaCy dependency parsing and visualize sentence structure.

2. Dataset Description

The dataset contains 15 sentences collected from news articles, technical texts, and everyday English sentences. These sentences include simple, compound, complex, interrogative, and passive voice structures.

3. Installation Steps

Install spaCy

Download language model

Load model in Python

4. Dependency Parsing Results

Show token tables extracted from sentences.

5. Tree Visualization

Include screenshots of dependency trees.

6. Sentence Structure Analysis

Explain subject, verb, object, and modifiers.

7. Comparison

Discuss differences between simple and complex sentences.

8. Conclusion

Syntactic analysis helps understand grammatical relationships and is important for many NLP applications.